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SEC : SR ELITE, AIIMS S60 & NEET MPL **DATE: 26-03-2022**

Sub: BOTANY **NEET GRAND TEST - 6** **Max.Marks: 720**

Guidelines:
In every subject:
a) In Section-A, 35 Questions will be given. Answer all 35 questions from Section-A
b) In Section-B, 15 questions will be given. Out of which answer 10 questions

BOTANY

SECTION –A

1. Kingdom plantae comprises all plants from various
   (1) Phyla
   (2) Sub species
   (3) Divisions
   (4) Domains
2. Cell wall is completely absent in
   (1) *Rhizobium*
   (2) *Chlamydomonas*
   (3) *Trichoderma*
   (4) Mycoplasma
3. **Incorrect** regarding prothallus of pteridophytes
   (1) Mostly photosynthetic
   (2) Multi cellular free living
   (3) Haploid
   (4) Bears roots
4. Bisexual monoecious gymnospermic plant is
   (1) *Pinus*
   (2) *Cycas*
   (3) *Cocos*
   (4) Maize
5. First organisms to colonise bare rock in xerosere are
   (1) Lichens
   (2) Pteridophytes
   (3) Some gymnosperms
   (4) Grasses
6. Which of the following has reserve food material which is very similar to amylopectin ?
   (1) *Ectocarpus*
   (2) *Gracilaria*
   (3) *Volvox*
   (4) *Ulothrix*
7. Match the following

A) Mono adelphous	I) China rose
B) Diadelphous	II) Pea
C) Polyadelphous	III) *Citrus*
D) Staminode	IV) Sterile stamen

	A	B	C	D
(1)	I	II	III	IV
(2)	II	III	I	IV
(3)	I	III	IV	II
(4)	II	III	IV	I

8. Obtaining sole rights for a innovation (or) novel creation of biological products is
   (1) Biopiracy
   (2) Biopatent
   (3) Bioremediation
   (4) Biofortification
9. Which of the following is adapter molecule ?
   (1) mRNA
   (2) rRNA
   (3) SnRNA
   (4) tRNA

10. Taq polymerase is

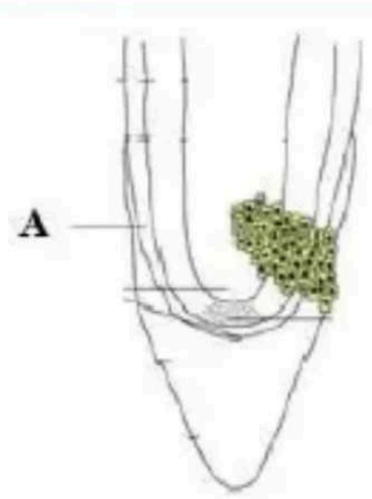
(1) DNA dependent RNA polymerase

(2) DNA dependent DNA polymerase

(3) RNA dependent DNA polymerase

(4) RNA dependent RNA polymerase

11. Identify "A" in the following diagram



(1) Central cylinder

(2) Protoderm

(3) Periblem

(4) Plerome

12. The term chromatin was coined by

(1) Robert Brown

(2) Robert Hooke

(3) Flemming

(4) R.Virchow

13. Cyanophycean granules is reserve food material in

(1) Protozoans

(2) Archaebacteria

(3) Cyanobacteria

(4) Protista

14. The following helps in DNA replication and distribution to daughter cells in bacteria

(1) Mesosome

(2) Slime layer

(3) Pili

(4) Nucleoid

15. Match the following

A) Mitochondria	I) Store oil
B) Amyloplasts	II) Store proteins
C) Aleuroplasts	III) Store starch
D) Elaioplasts	IV) Power Houses of the Cell

	A	B	C	D
(1)	I	II	III	IV
(2)	II	III	I	IV
(3)	IV	III	I	II
(4)	IV	III	II	I

16. Proteins encoded by gene *cry* I Ab control

(1) Corn borer

(2) Cotton boll worms

(3) Flies

(4) Mosquitoes

17. Carrier protein transporting glucose into cells is

(1) Aquaporins

(2) Porins

(3) GLUT-4

(4) Inulin

18. Dietary proteins are source of

(1) Non – essential aminoacids

(2) Essential aminoacids

(3) Non – essential lipids

(4) Essential fattyacids

19. In which succession climax reached faster?

(1) Succession on bare rock

(2) Succession on burned forest

(3) Succession on newly cooled lava

(4) All of these

20. Sister chromatids of a homologous chromosome move to opposite poles in

(1) Metaphase

(2) Telophase

(3) Anaphase II

(4) Anaphase I

21. Match the following

A) Diplotene	I) Terminalisation
B) Diakinesis	II) Chiasmata
C) Zygotene	III) Crossing over
D) Pachytene	IV) Synapsis

	A	B	C	D
(1)	III	IV	II	I
(2)	IV	II	III	I
(3)	II	IV	III	I
(4)	II	I	IV	III

22. Distribution of cell organelles occurs in
(1) S phase of interphase
(2) Cytokinesis
(3) Karyokinesis
(4) G_1 sub phase of interphase

23. Which of the following is a palindromic sequence ?
(1) 5'-CGTATG - 3'
(2) 5'-CGAATG-3'
(3) 5'GAATTC-3'
(4) 5'-GACTAC-3'

24. Guanine is
(1) Nucleoside
(2) Nitrogen base
(3) Nucleotide
(4) Nucleic acid

25. Identify the chemoautotrophs
(1) *Nitrobacter*
(2) *Nitrococcus*
(3) *Nitrosomonas*
(4) All the above

26. Unrelated to *Cladophora* is
(1) Belongs to Chlorophyceae
(2) Unicellular alga
(3) T.W Engelmann used in first action spectrum
(4) Flagellated gametes

27. The following are essential for reduction phase in Calvin cycle
(A) CO_2 (B) ATP
(C) $NADPH + H^+$ (D) O_2
(1) BC
(2) AB
(3) CD
(4) BD

28. Number of CO_2 molecules released during glycolysis of glucose molecule is
(1) 2
(2) 1
(3) 3
(4) zero

29. "Cells/tissues arising out of the same meristem have different structures at maturity" the above statement explains
(1) Heterophylly
(2) Plasticity
(3) Dedifferentiation is open
(4) Differentiation is open

30. Element which is not found in ferredoxin is
(1) Fe
(2) S
(3) Mg
(4) N

31. The term 'totipotency' refers to the capacity of a
(1) living plant cell to produce a whole plant
(2) scleried to generate whole plant
(3) seed to germinate
(4) cell to enlarge in size

32. The ploidy of perisperm is
(1) n
(2) 2n
(3) 3n
(4) 4n

33. Dry fruit is present in
(1) Mango
(2) Apple
(3) Coconut
(4) *Brassica*

34. Example for incomplete dominance is
(1) Human skin colour
(2) Height of pea plant
(3) Flower colour in snapdragon
(4) Shape of seed in pea

35. Who told for the first time that replication of DNA is semi conservative ?
(1) Meselson & Stahl
(2) Watson & Crick
(3) J.Cairans
(4) J.H.Taylor

SECTION – B

36. Unrelated to monocot stem is
(1) Vascular bundle is surrounded by sclerenchymatous bundle sheath
(2) Vascular bundles are conjoint and closed
(3) Water containing cavities are present within the vascular bundles
(4) Hypodermis is made up of living mechanical tissue

37.

Type / Nature of aminoacid	**Example**
A) Acidic	I) Tyrosine
B) Basic	II) Lysine
C) Neutral	III) Valine
D) Aromatic	IV) Glutamic acid

	A	**B**	**C**	**D**
(1)	III	IV	II	I
(2)	IV	II	III	I
(3)	II	IV	III	I
(4)	III	II	IV	I

38. Identify the **correct** statement from the following
(1) Acid insoluble pool roughly represents cytoplasmic composition
(2) Organelles are a part of acid soluble pool
(3) Most abundant chemical in living organisms are proteins
(4) Lipids are not strictly macromolecules

39. If a root cell has 24 chromosomes, then the number of chromosomes the first cell of male gametophyte of that plant is
(1) 24
(2) 6
(3) 12
(4) 18

40. Identify the **mismatch**
(1) Symport – Both molecules cross the membrane in the same direction
(2) Facilitated diffusion – uphill transport
(3) Antiport – Both molecules move in opposite directions
(4) Uniport – A molecule moves across a membrane independent of other molecules

41. **Match the Columns-I and II and identify** correct **match from the codes given below.**

Column-I	Column-II
A) Earthworm	I) Dioecious flowering plant
B) *Chara*	II) Bisexual animal
C) *Marchantia*	III) Dioecious bryophyte
D) Date palm	IV) Monoecious alga

Codes :

	A	**B**	**C**	**D**
(1)	II	IV	III	I
(2)	II	III	I	IV
(3)	III	IV	II	I
(4)	I	II	III	IV

42. Identify the correct statement regarding introns

(1) Are common in organisms which form polycistronic m RNA

(2) Are significant as they code for protein synthesis

(3) Do not undergo translation

(4) Get rejoined by DNA ligase during splicing

43. The number of amino acids getting linked into a polypeptide during translation following the mRNA with polypeptide sequence

5'----AUG CUC UUU UCU UAA ----3' will be

(1) Two

(2) Four

(3) Five

(4) One

44. During primary treatment, all solids that settle forms ____ A ___ and the supernatant forms ___B___

(1) A- Primary sludge; B –effluent

(2) A- Primary sludge; B – secondary effluent

(3) A- Activated sludge; B-clarified effluent

(4) A –Activated sludge; B-effluent

45. A foreign DNA fragment is attached to Cla I site of pBR 322. This recombinant vector is used to transform *Escherichia coli*. Then bacterial cells transferred on two different media, one containing ampicillin and the other containing tetracycline. The transformed cells containing the recombinant vector will

(1) Grow on both, tetracycline and ampicillin containing media

(2) Not grow on either tetracycline containing or ampicillin containing media

(3) Grow on tetracycline but not on ampicillin containing medium

(4) Grow on ampicillin but not on tetracycline containing medium

46. Select the mismatch pair

(1) Phenylalanine-UUU

(2) Serine – UCU

(3) Tryptophan – UGA

(4) Glycine – GGU

47. Statement – I : NADH is oxidised to NAD^+ rather slowly in fermentation, however the reaction is very vigorous in case of aerobic respiration

Statement – II : The NADH synthesized in glycolysis is transferred into the mitochondria and undergoes oxidative phosphorylation

(1) Both Statement I and Statement II are wrong

(2) Both Statement I and Statement II are true

(3) Statement I is correct, Statement II is wrong

(4) Both statement – I and statement – II are correct

48. Assertion (A) : In Angiosperms endosperm formation precedes embryo development

Reason (R) : In Angiosperms endosperm is post fertilization product and triploid

(1) A and R are true and R is the correct explanation of A.

(2) A and R are true and R is not the correct explanation of A.

(3) A is true, R is false.

(4) Both A and R are false.

49. Triple response test is shown by _____ phytohormone
(1) Auxins
(2) Gibberellins
(3) Cytokinins
(4) Ethylene

50. Cork cambium in dicot root arises from
(1) Endodermis
(2) Pericycle
(3) Conjuctive tissue
(4) Epidermis

ZOOLOGY

SECTION – A

51. Ascending limb of loop of Henle helps in
(1) Synthesis of urea
(2) Secretion of urea
(3) Secretion of Ca^{2+}
(4) Active transportation of glucose and amino acids

52. Choose the incorrect match.
(1) *Spongilla* – Extracellular digestion
(2) *Sepia* – Closed type of circulation
(3) *Ancylostoma* - Pseudocoelom
(4) *Locusta* – Open type of circulation

53. Regions with very high levels of species richness, high degree of endemism, accelerated habitat loss and require maximum protection are called
(1) Sacred groves
(2) National parks
(3) Biodiversity hotspots
(4) Zoological parks

54. Which of the following is true about thrombokinase?
(1) It activates heparin
(2) It activates fibrin to form as fibrinogen
(3) It activates prothrombin into thrombin
(4) It promotes formation of hydrogen bonds in the place of covalent bonds among fibrin threads

55. Biomass of tigers in a forest is
(1) Greater than the biomass of deer
(2) Greater than the biomass of producers
(3) Lesser than the biomass of deer
(4) Equal to the biomass of producers

56. Humification leads to accumulation of a dark coloured amorphous substance called humus that is
(1) Undergoing decomposition at an extremely fast rate
(2) Highly resistant to microbial action
(3) Degraded by some microbes and release of complex organic nutrients
(4) Degraded by detritivores into smaller organic particles

57. During embryonic development three primary germ layers are differentiated from
(1) Hypoblast
(2) Trophoblast
(3) Chorion
(4) Inner cell mass of embryo

58. The measure of population density of tigers in a national park is
(1) Indirect count
(2) Relative densities
(3) Biomass
(4) Sex ratio

59. The optic nerves leave the eye and the retinal blood vessels enter it at a point __A__ to and slightly __B__ the __C__ pole of the eye ball.
In the above, A, B and C respectively are
(1) Medial, below and posterior
(2) Anterior, below and posterior
(3) Medial, above and posterior
(4) Medial, above and anterior

60. In some children, ADA deficiency can be cured by

(1) Radiotherapy

(2) Giving nutrient supplement

(3) Bone marrow transplantation

(4) Antibiotics

61. Which of the following allows the desirable qualities of two different breeds to be combined ?

(1) Inbreeding

(2) Cross-breeding

(3) Species hybridisation

(4) Out-crossing

62. Statement-I : Radiation that is given off by nuclear waste is extremely damaging to organisms.

Statement-II : This radiation causes mutations at a very high rate

(1) Statement-I & II are true, Statement-II explains Statement-I

(2) Statement-I & II are true, Statement-II does not explain Statement-I

(3) Statement-I is true, but Statement-II is false

(4) Statement-I & II are false

63. Which of the following acts as a filter of the blood by trapping blood-borne microorganisms?

(1) Thymus

(2) Spleen

(3) Red bone marrow

(4) Epiphysis of bone

64. Formation of gametocytes of *Plasmodium* occurs in

(1) RBC of man

(2) Liver cells of man

(3) Gut wall of mosquito

(4) Lumen of gut of mosquito

65. Select the vector-borne helminth disease in the following

(1) Filariasis

(2) Pneumonia

(3) Chikungunya

(4) Malaria

66. Brain capacity of the first human-like being was

(1) 1400 cc

(2) 900 cc

(3) 1650 cc

(4) 650 – 800 cc

67. The process in which heritable variations enabling better survival are enabled to reproduce and leave greater number of progeny is called

(1) Natural selection

(2) Genetic drift

(3) Gene flow

(4) Mutation

68. Different structures evolving for the same function in

(1) Adaptive radiation

(2) Convergent evolution

(3) Divergent evolution

(4) Common ancestry

69. Among humans, how much percent of base sequence of the genome is the same?

(1) 99.9

(2) 50

(3) 0.01

(4) 2

70. The affected individual with trisomy of 23 (Karyotype of 47, XXY) shows all the following features except

(1) Congenital heart disease

(2) Sterility

(3) Tall stature and gynaecomastia

(4) Overall masculine development

71. In cockroach, which of the following is related to the Gizzard?
(1) It is the main site of absorption of nutrients
(2) It secretes amylases, proteases and lipases
(3) It is lined by only endoderm
(4) It has an outer layer of thick circular muscles and thick inner cuticle forming six highly chitinous plates called teeth

72. Oral contraceptive pills of progestogen - estrogen combinations do not affect
(1) Spermiogenesis
(2) Entry of sperms into uterus
(3) Ovulation
(4) Quality of cervical mucus

73. Find out the steroid hormone which is in more amounts by the end of follicular phase of menstrual cycle
(1) Progesterone
(2) LH
(3) Inhibin
(4) Estrogen

74. In human, secondary spermatocytes are formed by
(1) Second maturation division
(2) Meiosis – II
(3) Reduction division
(4) Mitosis

75. In which of the following sets, one animal species benefited and the other animal species is neither harmed nor benefited?
(1) Dog – Copepod
(2) Orchid – Bumblebee
(3) Cuscuta – Hedge plant
(4) Sea anemone – Clown fish

76. Hormone that binds with intracellular receptor is
(1) Adrenaline
(2) Aldosterone
(3) Oxytocin
(4) FSH

77. Which of the following is not related to cortisol ?
(1) It inhibits gluconeogenesis
(2) It stimulates proteolysis
(3) It stimulates RBC production
(4) It suppresses the immune response

78. Hepatitis B vaccine is
(1) Toxoid vaccine
(2) Inactivated whole agent vaccine
(3) Produced from yeast
(4) Attenuated whole agent vaccine

79. Repolarisation of an axon occurs mainly due to
(1) Influx to K^+
(2) Out flux of K^+
(3) Out flux of Na^+
(4) Influx of Na^+

80. Glisson's capsule
(1) Surrounds glomerulus
(2) Is made of hepatocytes
(3) Covers adrenal gland
(4) Covers hepatic lobules

81. Choose the oviparous homeotherms
(1) *Pterophyllum* and *Aptenodytes*
(2) *Ichthyophis* and *Macropus*
(3) Platypus and *Pteropus*
(4) *Ornithorhynchus* and *Struthio*

82. Match the following

A) Tendons and ligaments	i) Areolar connective tissue
B) Dermis of skin	ii) Dense irregular tissue
C) Subcutaneous layer	iii) Areolar and adipose connective tissue
D) Supporting framework for epithelium	iv) Dense regular connective tissue

	A	B	C	D
(1)	ii	iv	i	iii
(2)	iv	ii	i	iii
(3)	i	ii	iv	iii
(4)	iv	ii	iii	i

83. The difference between pO_2 of blood in pulmonary vein and pO_2 of blood in pulmonary artery is about

(1) 45 mm Hg

(2) 40 mm Hg

(3) 64 mm Hg

(4) 55 mm Hg

84. In a skeletal myofibril, thin filaments are found in

(1) Only A – band

(2) Only I – band

(3) H – zone

(4) A-band and I-band

85. Find out the movable unpaired bone in human body

(1) Incus

(2) Scapula

(3) Sphenoid

(4) Mandible

SECTION – B

86. Which of the following is not a consequence of deforestation?

(1) Desertification

(2) Soil erosion

(3) Reduction in CO_2 pollution in the atmosphere

(4) Disturbs hydrological cycle

87. Study the following statements.

a) Specialised skeletal connective tissues include cartilage and bone.

b) In all connective tissues, the cells secrete fibres and these cells also secrete modified polysaccharides

c) Compound epithelium has a limited role in secretion and absorption.

d) Salivary glands are the multicellular glands.

In the above,

(1) All are true except b

(2) All are false except c

(3) All are true except a

(4) All are true except d

88. Statement – I : Strength of ventricular contraction is increased by sympathetic division of ANS.

Statement – II : Cardiac output is increased by catecholamines.

(1) Both statement I and II are correct

(2) Statement I is incorrect, II is correct

(3) Statement I is correct, II is incorrect

(4) Both statement I and II are wrong

89. A stable biotic community should not show

(1) Resistant or resilient to occassional disturbances

(2) Resistant to invasions by alien species

(3) Less year-to-year variation in total biomass

(4) Too much variation in productivity from year to year

90. *Psilophyton* is the common ancestor for
(1) Seed ferns, conifers and bryophytes
(2) Gymnosperms, ferns and horsetails
(3) Cycads, gnetales and lycopods
(4) Sphenopsids, cycads and bryophytes

91. Kyoto Protocol was signed in 1997 to
(1) Protect aquatic species from the biomagnification
(2) Protect earth from global warming
(3) Protect wetlands
(4) Control the emission of ozone depleting substances

92. Study the following statements regarding cockroach.
a) A pair of vas deferens open into ejaculatory duct through seminal vesicle.
b) The sperms are stored in the seminal vesicles and are glued together in the form of bundles called spermatophores.
c) Layers of spermatophore capsule are formed by the secretions of utriculi majors, ejaculatory duct and phallic gland.
d) A pair of spermathecae in female store fertilized eggs.
Which of the above are correct?
(1) c and d
(2) a, c and d
(3) a, b and c
(4) b, c and d

93. GIFT is attempted to female who
(1) Can not provide suitable environment for fertilization
(2) Can not produce ovum
(3) Can not provide suitable environment for implantation
(4) Is having normal ovarian cycles

94. In human, contraction of internal intercostal muscles causes
(1) Decreased intrapulmonary pressure
(2) Increased pulmonary volume
(3) Increased thoracic volume
(4) Increased intrapulmonary pressure

95. Secretion of ANF from the atria of the heart results in
(1) Increased BP
(2) Vasoconstriction
(3) Increased reabsorption of Na^+ and water
(4) Vasodilation

96. A large quantity of urban sewage is drained to a nearby river. Which among the given conditions would happen after mixing of sewage into the river ?
a. BOD of receiving water body increases
b. DO of receiving water body decreases
c. It will not cause mortality among fishes and other aquatic creatures
d. It will lead to nutrient enrichment of receiving water body
(1) a, b and c
(2) a, b and d
(3) b and c
(4) c and d

97. Find out the correct inter chain S-S linkage in human insulin
(1) A19 – B20
(2) A7 – B6
(3) A6 – A11
(4) A20 – B19

98. The structural differences in genitalia of individuals belonging to different animal species interfere with mating, it is called
(1) Mechanical isolation
(2) Gametic isolation
(3) Behavioural isolation
(4) Temporal isolation

99. Statement-I : The epithelial cells of Bowman's capsule called podocytes are arranged in an intricate manner so as to leave some minute spaces called filtration slits or slit pores.

Statement-II : The first step of urine formation (glomerular filtration) is a non-selective process.

(1) Statement-I & II are true

(2) Statement-I & II are false

(3) Statement-I is true, but Statement-II is false

(4) Statement-I is false, but Statement-II is true

100. Desert lizards bask in the Sun and absorb heat when their body temperature drops below the comfort zone, this is

(1) Morphological adaptation

(2) Physiological adaptation

(3) Behavioural adaptation

(4) Anatomical adaptation

PHYSICS

SECTION – A

101. Two trains travelling at 12 ms^{-1} and the other at 20 ms^{-1} are headed towards one another along a straight track. Both the drivers apply brakes simultaneously when they are 500m apart. If each train has a retardation of $1ms^{-2}$, the separation after they stop is

(1) 192.5m

(2) 425m

(3) 244m

(4) 355.5m

102. A boat which can move with a speed of 5m/s relative to water crosses a river of width 480m flowing with a constant speed of 4m/s. What is the time taken by the boat to cross the river along the shortest path.

(1) 96s

(2) 160s

(3) 240s

(4) 120s

103. A circular platform of radius 'r' is rotating at frequency 'n'. A small coin kept at a distance equal to half of its radius is ready to slip. The same coin will just slip, when it is at a distance of '2r/3' from the centre on another platform of radius '2r', rotating at frequency '3n'.Then ratio between coefficients of friction in two cases is

(1) 12:1

(2) 1:9

(3) 9:1

(4) 1:12

104. Assertion (A) : A player lower his hands while catching a cricket ball to reduce impulsive force.

Reason (R) : With increase in time of action, the impulsive force increases.

(1) A and R are true and R is the correct explanation of A

(2) A and R are true and R is not the correct explanation of A

(3) A is true, R is false

(4) A is false, R is true

105. A lift carries a person with a weighing machine in it up and down with the same constant acceleration. The weights recorded by the machine are in the ratio 5:3. The acceleration of the lift is

(1) $2.45 ms^{-2}$

(2) $3.33 ms^{-2}$

(3) $2.67 ms^{-2}$

(4) 4.95 ms^{-2}

106. A uniform disc of mass 300 kg is rotating freely about a vertical axis through its centre at a constant angular velocity ω_0. A boy of mass 30 kg starts from the centre and moves along a radius to the edge of the disc. Angular velocity of the disc will now become

(1) $\frac{\omega_0}{6}$

(2) $\frac{\omega_0}{5}$

(3) $\frac{4\omega_0}{6}$

(4) $\frac{5\omega_0}{6}$

107. A disc at rest at the top of the inclined plane of height 'h' rolls down without slipping and acquires a velocity 'v' on reaching the bottom. If the same disc slides down a smooth incline and acquires the same velocity on reaching the bottom then the height of smooth incline is

(1) h/3

(2) h/2

(3) $\frac{2h}{3}$

(4) h

108. An accurate pendulum clock is mounted on ground floor of a high building. How much time it lose or gain in one day if it is transferred to stop story of a building which is 200 m higher than the ground floor? (Radius of earth is 6.4×10^6 m).

(1) it will lose 6.2 s

(2) it will lose 2.7 s

(3) it will gain 5.2 s

(4) it will gain 1.6 s

109. Two black bodies P and Q are emitting radiations with maximum intensities at wavelengths in the ratio 1 : 2. The total energies radiated per second per unit area by them will be in the ratio of

(1) 2 : 1

(2) 4 : 1

(3) 16 : 1

(4) 1 : 16

110. A material has Poisson's ratio 0.5. If a uniform rod of same material suffers a longitudinal strain of 2×10^{-3}, what is the percentage increase in its volume?

(1) 2%

(2) 4%

(3) 0%

(4) 5%

111. When a capillary tube of radius 2mm is placed in a liquid, the mass of liquid raised is found to be 2g. If the angle of contact between the material of tube and liquid is 37^0 and density of liquid is $0.8gcc^{-1}$ then surface tension of the liquid is nearly

(1) 1.5 N/m

(2) 2N/m

(3) 3 N/m

(4) 4 N/m

112. A kerosene tank has kerosene up to a height of 4 m. There is a hole of 1 cm^2 cross-section at the bottom through which it leaks. The initial leak will be at the rate of V cc/min. After half of the tank is emptied the rate of leak of kerosene from tank in cc/min will be

(1) V

(2) V/2

(3) 3V

(4) $\frac{V}{\sqrt{2}}$

113. A container has N molecules at absolute temperature T. If the number of molecules is doubled but kinetic energy of gas in the box remains the same as before, the absolute temperature of the gas is
(1) T
(2) T/2
(3) 2T
(4) zero

114. A particle of mass 10 g is executing simple harmonic motion with an amplitude of 0.5 m and periodic time of (π/5) seconds. The maximum value of the force acting on the particle is
(1) 25 N
(2) 5 N
(3) 2.5 N
(4) 0.5 N

115. The linear density of a vibrating string is 10^{-4} kg/m. A transverse wave is propagating on the string, which is described by the equation y=0.02 sin (x + 30t) where x and y are in meters and the time t is in seconds. The tension in the string is
(1) 0.09 N
(2) 0.36 N
(3) 0.9 N
(4) 3.6 N

116. The angle of polarization of material of a transparent slab is 60^0. If a light ray is incident on the slab at a glancing angle 30^0 then the angle of deviation of reflected ray is
(1) 30^0
(2) 60^0
(3) 45^0
(4) 0^0

117. An astronomical telescope of tenfold angular magnification has a length of 44cm. The focal length of the objective is
(1) 4 cm
(2) 40 cm
(3) 44 cm
(4) 440 cm

118. In a double slit experiment, interference is obtained from electron waves produced in an electron gun supplied with voltage V. If λ is wavelength of the beam, D is the distance of screen, d is the spacing between coherent source, h is Planck's constant, e is charge on electron and m is mass of electron, then fringe width is given as
(1) $\frac{hD}{\sqrt{2meV}\,d}$
(2) $\frac{2hD}{\sqrt{meV}\,d}$
(3) $\frac{hd}{\sqrt{2meV}\,D}$
(4) $\frac{2hd}{\sqrt{meV}\,D}$

119. If in the experiment of diffraction at a single slit, the width of slit is equal to twice the wavelength of light used, the angle of diffraction for the first minimum will be
(1) 30°
(2) 45°
(3) 60°
(4) 90°

120. A hollow insulated conducting sphere of radius 20 cm is given a charge q. The electric field at a point 10 cm from the surface of the sphere is 50 V/m. The electric potential at a distance 10 cm from center of the sphere is
(1) zero
(2) 45V
(3) 22.5V
(4) 90V

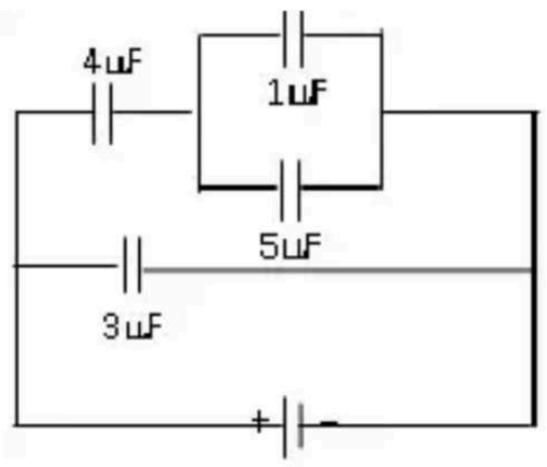
121. Match the following

A. Radio waves	p. green house effect
B. Micro waves	q. production of melanin
C. Infrared waves	r. communication in cellular phones
D. Ultra violet rays	s. raise of temperature of food containg water

(1) A-p; B-q; C-r; D-s
(2) A-r; B-s; C-p; D-q
(3) A-r; B-p; C-s; D-q
(4) A-q; B-p; C-s; D-r

122. A parallel plate capacitor with the distance between the plates d is charged to potential V and disconnected from the battery. The intensity of electric field between the plates is E. Now if the distance between the plates is increased to 2d, the electric field will be
(1) E
(2) 2E
(3) E/2
(4) nothing can be predicted

123. For the circuit shown in the adjoining figure, if the charge on 4μF capacitor is 24μC then energy stored in 3μF capacitor is



(1) 300μJ
(2) 150μJ
(3) 200μJ
(4) 100μJ

124. A body suspended in water with the help of a string. Tension in the string when water is at 4^0C is T. The tension in the string when the temperature of water is changed is (Ignore thermal expansion of the body)
(1) > T from 0^oC to 4^oC only
(2) always T
(3) <T for any temperature other than 4^oC
(4) >T for any temperature other than 4^oC

125. A beam of ions with velocity 2×10^5 ms^{-1} enters normally into a uniform magnetic field of 4×10^{-2} T. If the specific charge of the ions is 5×10^7 C kg^{-1}, the radius of the circular path described will be
(1) 0.10 m
(2) 0.16 m
(3) 0.20 m
(4) 0.25 m

126. A current of 2 ampere produces a deflection of 30° in a tangent galvanometer. A deflection of 60° will be produced in it by a current of
(1) 1A
(2) 3A
(3) 4A
(4) 6A

127. The power factor for series L-C-R circuit at resonance is
(1) 1
(2) $\frac{1}{2}$
(3) $\frac{1}{\sqrt{2}}$
(4) 0

128. A milliammeter of range 10 mA has a coil of resistance 1Ω. To use it as a voltmeter of range 10 V, the resistance that must be connected in with it is
(1) 9Ω in series
(2) 99Ω in parallel
(3) 999Ω in series
(4) 999Ω in parallel

129. The average e.m.f. induced in a coil when a current changes from 0 to 2A in 0.05 sec is 8V. If a current of 0.5 A flows through the coil then energy stored in the coil is

(1) 25 mJ

(2) 50 mJ

(3) 12.5 mJ

(4) 100 mJ

130. A capacitor, an inductor and an electric bulb are connected in series to an AC supply of variable frequency. If the frequency of the supply is increased gradually, then the electric bulb is found to

(1) increase in brightness

(2) decrease in brightness

(3) increase, reach a maximum and then decrease in brightness

(4) show no change in brightness

131. Which of the following statements is correct?

(1) The stopping potential increases with increasing intensity of incident light

(2) The saturated photocurrent increases with increasing intensity of light

(3) The saturated photocurrent is proportional to applied voltage

(4) The current in a photocell increases with increasing frequency of light

132. The energy that should be added to an electron to reduce its de–Broglie wavelength from 1nm to 0.5 nm is

(1) four times the initial energy

(2) equal to the initial energy

(3) twice the initial energy

(4) thrice the initial energy

133. According to Bohr's theory of H-atom, the product of Binding energy of electron and radius in the nth orbit is

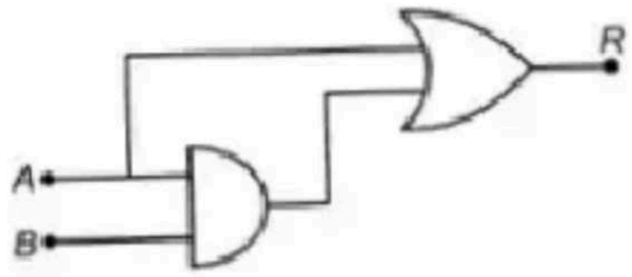
(1) Proportional to n^2

(2) Inversely proportional to n^2

(3) Independent of n

(4) None

134. In the combination of logic gates shown in the circuit. If A is at 0 V and B is at 5 V, then the potential of R is



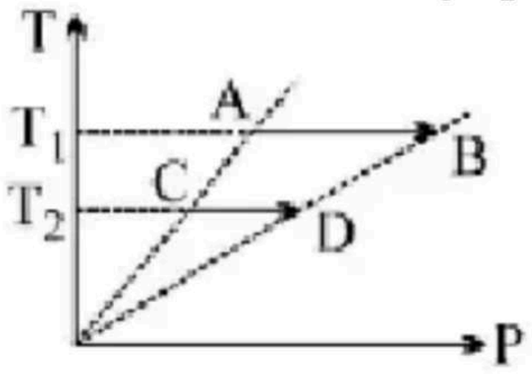
(1) 0 V

(2) 5 V

(3) 10 V

(4) 20 V

135. On a T-P diagram, two moles of ideal gas perform process AB and CD. If the work done by the gas in the process AB is two times the work done in the process CD then what is the value of T_1/T_2 ?



(1) 0.5

(2) 1

(3) 2

(4) 4

SECTION – B

136. A body projected at an angle other than 90^0 rises to a height of 40m in 2sec. Then the additional time it takes before reaching the ground is ($g = 10\ ms^{-2}$)

(1) 6s

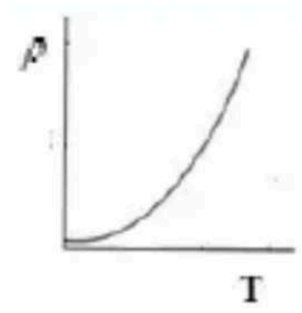
(2) 4s

(3) 2s

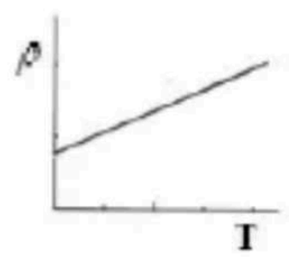
(4) 3s

137. Variation of resistivity of nichrome as a function of absolute temperature is

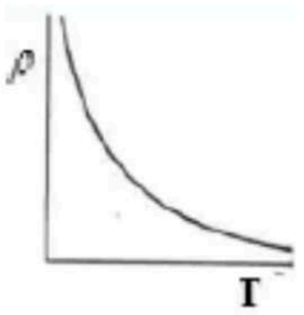
(1)



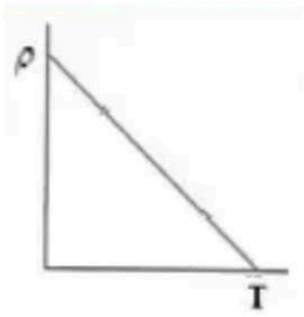
(2)



(3)



(4)



138. The distance covered by a body to come to rest when it is moving with a speed of 4 ms^{-1} is 's' when a retarding force F is applied. If the KE is doubled, the distance covered by it to come to rest for the same retarding force F is

(1) 4s

(2) 6s

(3) 2s

(4) 8s

139. In which of the following cases the work done increases the potential energy?

(1) Both conservative and non-conservative forces

(2) Conservative force only

(3) Non-conservative force only

(4) Neither conservative nor non-conservative forces

140. An ice block of mass 42 kg is dragged on a horizontal surface through a distance of 10m. Coefficient of friction between the surfaces is 0.2. If all the heat produced is utilized for melting of ice, the mass of ice melted is nearly

(1) 2.5 kg

(2) 2.5 g

(3) 2.5 mg

(4) 25 kg

141. The length of a potentiometer wires is l. A cell of emf E is balanced at a length $\left(\frac{l}{3}\right)$ from positive end of the wire. If the length of the wire is increased by $\left(\frac{l}{2}\right)$, the distance at which the same cell gives the balancing point is (Cell in the primary is ideal and no series resistance is present in the primary circuit)

(1) $\frac{4l}{2}$

(2) $\frac{l}{3}$

(3) $\frac{2l}{3}$

(4) $\frac{l}{2}$

142. An alpha-particle is projected towards a Cu nucleus, an Au nucleus and an Ag nucleus with same speed from a large distance. The distances of closest approaches were found to be d_1, d_2 and d_3 respectively. Which of the following is correct?

(1) $d_1 < d_2 < d_3$

(2) $d_1 < d_3 < d_2$

(3) $d_2 < d_3 < d_1$

(4) $d_3 < d_1 < d_2$

143. A thin convergent glass lens $\left(\mu_g = 1.5\right)$ has a power of +5.0 D. When this lens is immersed in a liquid of refractive index μ_l it acts as a divergent lens of focal length 100 cm. The value of μ_l must be

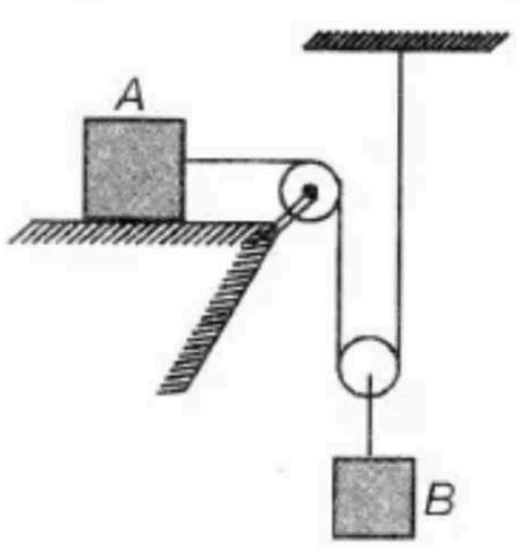
(1) 4/3

(2) 5/3

(3) 4/5

(4) 3/5

144. If the work done by the string on block A is W, then work done by the string on the block B will be



(1) –W

(2) $-\frac{W}{2}$

(3) 2W

(4) Zero

145. Two particles of masses 2kg and 3kg initially at rest. They are moving towards each other under mutual gravitational force of attraction. If at an instant their velocities are V_1 and V_2 respectively then the velocity of their centre of mass at that instant is

(1) $2V_1 + 3V_2$

(2) $\frac{3V_1 - 2V_2}{5}$

(3) zero

(4) $\frac{3V_1 + 2V_2}{5}$

146. The binding energy per nucleon for C^{12} is 7.68 MeV and that for C^{13} is 7.47 MeV. The energy required to remove a neutron from C^{13} is

(1) 0.21 MeV

(2) 2.52 MeV

(3) 4.95 MeV

(4) 2.75 MeV

147. The time period of vibration of a dip needle when box of dip circle is perpendicular to magnetic meridian is 1 second. When same magnetic needle is made to vibrate in the horizontal plane, the time period becomes $3^{1/4}$ second. Then the angle of dip is

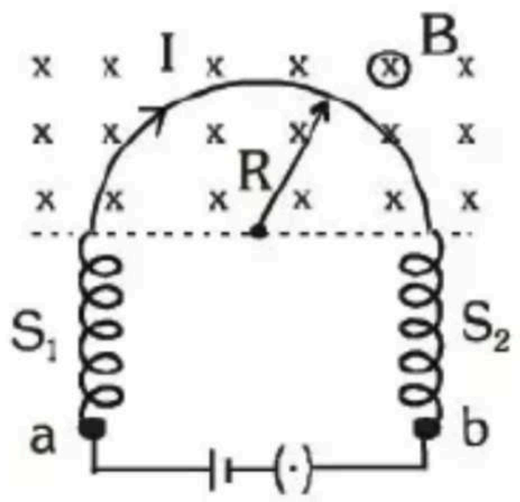
(1) 30^0

(2) 45^0

(3) 60^0

(4) 90^0

148. A wire in the form of semi circle lies on a frictionless table. The ends of semi circle are attached to identical springs S_1 and S_2 whose other ends a and b are fixed. If the current in the wire is I then tension in each spring is



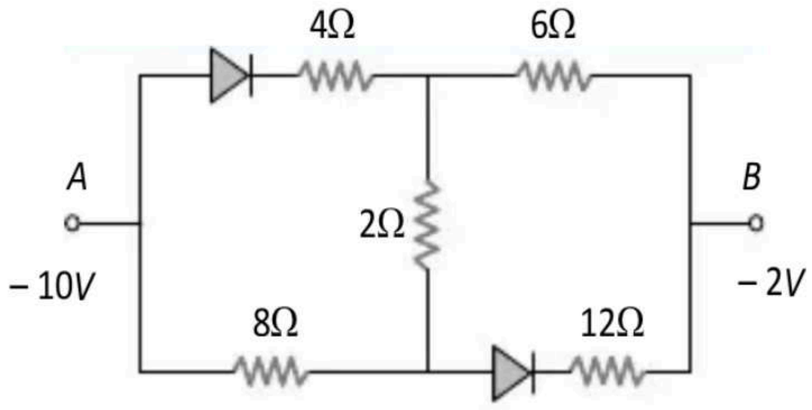
(1) πRIB

(2) 2RIB

(3) IRB

(4) Zero

149. In the following circuit the equivalent resistance between A and B is



(1) $\frac{20}{3}\Omega$

(2) $10\ \Omega$

(3) $16\ \Omega$

(4) $20\ \Omega$

150. The sum and difference of the surface areas of two soap bubbles are in the ratio 5 : 3. The ratio of the excess pressures in them is

(1) $\sqrt{3}:\sqrt{5}$

(2) $\sqrt{5}:\sqrt{3}$

(3) 1 : 4

(4) 1 : 2

CHEMISTRY
SECTION – A

151. Which of the following reaction is not correct according to the law of conservation of mass?

(1) $2Mg(s)+O_2(g)\longrightarrow 2MgO(s)$

(2) $C_3H_8(g)+O_2(g)\longrightarrow CO_2(g)+H_2O(g)$

(3) $P_4(s)+5O_2(g)\longrightarrow P_4O_{10}(s)$

(4) $CH_4(g)+2O_2(g)\longrightarrow CO_2(g)+2H_2O(g)$

152. In which of the following molecule/ion all the bonds are not equal?

(1) XeF_4

(2) BF_4^-

(3) C_2H_4

(4) SiF_4

153. The ratio of the kinetic energies of equal number of moles of H_2 and He at the same temperature is

(1) 1 : 2

(2) 2 : 1

(3) 1 : 1

(4) 4 : 1

154. $N_2+3H_2\longrightarrow 2NH_3\ ;\ \Delta H=-46\,\text{K.Cals}.$

From the above reaction, heat of formation of ammonia is

(1) 46 K.Cals

(2) – 46 K.Cals

(3) – 23 K.Cals

(4) 23 K.Cals

155. Consider the reaction,

$N_2(g)+3H_2(g)\rightleftharpoons 2NH_3(g)$

Which of the following is correct, if the total pressure at which the equilibrium is established, is increased without changing the temperature?

(1) K will remain same

(2) K will decrease

(3) K will increase

(4) K will increase initially and decreases when pressure is very high

156. Which of the following will produce a buffer solution when mixed in equal volumes?

(1) $0.1 mol\ dm^{-3}\ NH_4OH$ and $0.1 mol\ dm^{-3} HCl$

(2) $0.05 mol\ dm^{-3}\ NH_4OH$ and $0.1 mol\ dm^{-3} HCl$

(3) $0.1 mol\ dm^{-3}\ NH_4OH$ and $0.05 mol\ dm^{-3} HCl$

(4) $0.1 mol\ dm^{-3} CH_4COONa$ and $0.1\ mol\ dm^{-3} NaOH$

157. Which of the following elements does not show disproportionation tendency?

(1) Cl

(2) Br

(3) F

(4) I

158. How many unit cells are present in 5.0 gm of crystal AB (formula mass AB = 40) having rock salt type structure? (N_A = Avogadro's no)

(1) N_A

(2) $\frac{N_A}{10}$

(3) $4N_A$

(4) $\frac{N_A}{32}$

159. Which of the following statements is correct regarding a solution of two components A and B exhibiting positive deviation from ideal behaviour?

(1) Intermolecular attractive forces between A-A and B-B are stronger than those between A-B.

(2) Δ_{mix} H = 0 at constant T and P.

(3) Δ_{mix} V = 0 at constant T and P.

(4) Intermolecular attractive forces between A-A and B-B are equal to those between A-B.

160. How many grams of cobalt metal will be deposited when a solution of cobalt(II) chloride is electrolyzed with a current of 10 amperes for 109 minutes? (1 Faraday = 96,500 C; Atomic mass of Co = 59 u)

(1) 4.0

(2) 20.0

(3) 40.0

(4) 0.66

161. The activation energy of a reaction can be determined from the slope of which of the following graphs?

(1) $\ln k \ vs \frac{1}{T}$

(2) $\frac{T}{\ln k} vs \frac{1}{T}$

(3) $\ln k \ vs \ T$

(4) $\frac{\ln k}{T} vs T$

162. Which property of colloidal solution is independent of charge on the colloidal particles?

(1) Electroosmosis

(2) Tyndall effect

(3) Coagulation

(4) Electrophoresis

163. On which of the following properties does the coagulating power of an ion depend?

(1) The magnitude of the charge on the ion alone

(2) Size of the ion alone

(3) Both magnitude and sign of the charge on the ion

(4) The sign of charge on the ion alone

164. A mineral usually has large amount of undesirable impurities. These impurities are called

(1) Matrix or gangue

(2) Slag

(3) Flux

(4) Ore

165. Extraction of gold and silver involves leaching with CN^- ion. Silver is later recovered by

(1) Distillation

(2) Zone refining

(3) Displacement with Zn

(4) Liquation.

166. Which of the following is a cationic detergent

(1) Sodium lauryl sulphate

(2) Sodium steartate

(3) Cetyl trimethyl ammonium bromide

(4) Sodium dodecyl benzene sulphonate

167. In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it

(1) $CO_2 < SiO_2 < SnO_2 < PbO_2 -$

Increasing oxidising power

(2) $HF < HCl < HBr < HI$ – Increasing acidic strength

(3) $H_2O < H_2S < H_2Se < H_2Te$ –

Increasing pKa value

(4) $NH_3 < PH_3 < AsH_3 < SbH_3$ –

Increasing reducing power

168. Match list – I with list – II

List – I		List – II	
A)	$ClONO_2 + HCl \rightarrow Cl_2 + HNO_3$	i)	Acid rain
B)	$NO_2(g) \xrightarrow{h\upsilon} NO(g) + O(g)$	ii)	Environmental friendly reaction
C)	$4NO_2(g) + O_2(g) + 2H_2O(l) \rightarrow 4HNO_3$	iii)	Photo chemical smog
D)	$H_2C = CH_2 + O_2 \xrightarrow{Pd} CH_3CHO$	iv)	Ozone hole

Choose the correct answer from the options given below

(1) A – (iii), B – (ii), C – (iv), D – (i)

(2) A – (i), B – (ii), C – (iii), D – (iv)

(3) A – (ii), B – (iii), C – (iv), D – (i)

(4) A – (iv), B – (iii), C – (i), D – (ii)

169. In the kjeldahl's method for estimation of nitrogen present in a sample ammonia evolved from 0.75g of sample neutralized 10 ml of 1 M H_2SO_4. The percentage of nitrogen in the sample is

(1) 37.33

(2) 45.33

(3) 35.33

(4) 18.66

170. $A - B \longrightarrow A^{\bullet} + B^{\bullet}$; $\Delta H = + x\,kJ\,mol^{-1}$

$A - B \longrightarrow A^{+} + B^{-}$; $\Delta H = + y\,kJ\,mol^{-1}$

The relationship between x and y in gaseous phase is

(1) $x = y$

(2) $x > y$

(3) $x < y$

(4) $x = 2y$

171. To prepare 3-ethyl pentan – 3 – ol, the reagents required are

(1) $CH_3CH_2MgBr + CH_3COCH_2CH_3 / H_3O^+$

(2) $CH_3MgBr + (CH_3CH_2)CO / H_3O^+$

(3) $CH_3CH_2MgBr + (CH_3CH_2)CO / H_3O^+$

(4) $CH_3CH_2CH_2MgBr + CH_3COC_2H_5 / H_3O^+$

172. Which one of the following reaction does not occur when heated with copper at 300°C

(1) $(CH_3)_2 CHOH \rightarrow (CH_3)_2 C = O$

(2) $(CH_3)_3 C - OH \rightarrow (CH_3)_2 C = CH_2$

(3) $CH_3CH_2CH_2OH \rightarrow CH_3CH_2CHO$

(4) $C_6H_5OH \rightarrow C_6H_6$

173. $C_3H_8O(X) \xrightarrow[H_2SO_4]{K_2Cr_2O_7} C_3H_6O \xrightarrow[Warm]{I_2+NaOH} CHI_3$

"X" can be

(1) $CH_3CH_2CH_2OH$

(2) $CH_3CHOHCH_3$

(3) $CH_3OCH_2CH_3$

(4) CH_3CH_2CHO

174. Gabriel phtalimide reaction is useful for preparation of

(1) $C_6H_5NH_2$ and $C_2H_5NH_2$

(2) $C_2H_5NH_2$ and $C_3H_7NH_2$

(3) $C_2H_5NH_2$ and $(C_2H_5)_2NH$

(4) $C_6H_5CH_2NH_2$ and $C_6H_5NH_2$

175. The correct order of magnetic moments (Spin only values in B.M) among the following

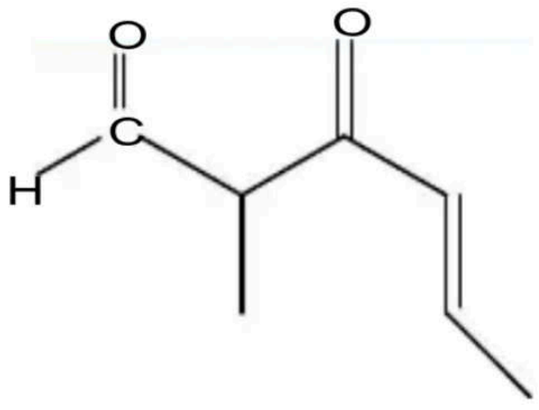
(1) $[Fe(CN)_6]^{4-} > [MnCl_4]^{2-} > [COCl_4]^{2-}$

(2) $[MnCl_4]^{2-} > [Fe(CN)_6]^{4-} > [COCl_4]^{2-}$

(3) $[MnCl_4]^{2-} > [COCl_4]^{2-} > [Fe(CN)_6]^{4-}$

(4) $[Fe(CN)_6]^{4-} > [COCl_4]^{2-} > [MnCl_4]^{2-}$

176. The IUPAC name of the compound



is

(1) 5 – formyl hex – 2 – en – 3 – one

(2) 5 – methyl – 4 – oxo hex – 2 – en – 5 – al

(3) 3 – keto – 2 – methyl hex – 5 – enal

(4) 3 – keto – 2 – methyl hex – 4 – enal

177. In which of the following reactions Benzene will not be formed

(1) $C_6H_5COONa \xrightarrow{NaOH+CaO}$

(2) $C_6H_5OH \xrightarrow{Zn/\Delta}$

(3) $C_6H_5N_2Cl \xrightarrow{H_3PO_2}$

(4) n – hexane $\xrightarrow[AlCl_3/300^\circ+HCl]{Anhydrons}$

178. $X \xrightarrow{\text{Soda lime}} Y \xrightarrow{Cl_2/\text{uv-light}} Z \xrightarrow[\text{dryether}]{2\,Na}$

$CH_3-CH_2-CH_2-CH_3 + 2\,NaCl$. Then X, Y and Z are

(1) Sodium ethanoate, ethane and ethyl chloride

(2) Sodium propanoate, ethane and methyl chloride

(3) Sodium butanoate, ethane and ethyl chloride

(4) Sodium propanoate, ethane and ethyl chloride

179. Wrong statement about solvay process

(1) Byeproduct is $CaCl_2$

(2) Raw materials are NaCl, NH_3 and limestone

(3) NH_3 is recovered by using $Ca(OH)_2$

(4) K_2CO_3 can be prepared by this method

180. Which of the following is incorrect statement

(1) Aqueous solution of Boric acid – Mild antiseptic

(2) Aqueous solution of Borax – alkaline

(3) Borax is used in pyrex glass

(4) Aqueous solution of $B(OH)_3$ is a Bronsted acid

181. Identify incorrect statement among the following

(1) CO is a lewis base (or ligand)

(2) Bond length in CO_2 is 115 pm due to resonance

(3) Diamond is used in the making of dyes

(4) Graphite is an insulator

182. The glycosidic linkage involved in linking the glucose units in amylase part of starch is

(1) $C_1 - C_4$ β – linkage

(2) $C_1 - C_6$ α – linkage

(3) $C_1 - C_6$ β – linkage

(4) $C_1 - C_4$ α – linkage

183. Which one of the following is not correctly represented regarding polymers

(1) Bakelite – Cross linked polymer

(2) Nylon–2– nylon–6 – Biodegradable polymer

(3) Neoprene – Condensation polymer

(4) Caprolactam – Nylon - 6

184. Identify the incorrect match

Name		IUPAC official name	
A)	Unnilunium	i)	Mendelevium
B)	Unniltrium	ii)	Lawrencium
C)	Unnilhexium	iii)	Sea borgium
D)	Unununnium	iv)	Darmstadtium

(1) (A) (i)

(2) (B) (ii)

(3) (C) (iii)

(4) (D) (iv)

185. In which pair of ions both the species contain S – S bond

(1) $S_4O_6^{2-}, S_2O_3^{2-}$

(2) $S_2O_7^{2-}, S_2O_8^{2-}$

(3) $S_4O_6^{2-}, S_2O_7^{2-}$

(4) $S_2O_7^{2-}, S_2O_3^{2-}$

SECTION – B

186. The quantum number not obtained from the Schrodinger's wave equation is

(1) n

(2) l

(3) m

(4) s

187. PCl_5, PCl_3, and Cl_2 are at equilibrium at 500 K in a closed container and their concentrations are $0.8\times10^{-3} mol\,L^{-1}$, $1.2\times10^{-3} mol\,L^{-1}$ and $1.2\times10^{-3} mol\,L^{-1}$, respectively. The value of K_c for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ will be

(1) $1.8\times10^{3} mol\,L^{-1}$

(2) $1.8\times10^{-3} mol\,L^{-1}$

(3) $1.8\times10^{-3} mol^{-1}\,L$

(4) $0.55\times10^{4} mol^{-1}L$

188. Lithium has a bcc structure. Its density is $530\ kg\,m^{-3}$ and its atomic mass is $6.94\ g\,mol^{-1}$. Calculate the edge length of a unit cell of lithium metal. $\left(N_A = 6.02\times10^{23} mol^{-1}\right)$

(1) 527 pm

(2) 264 pm

(3) 154 pm

(4) 352 pm

189. In the electrochemical cell :

$Zn|ZnSO_4(0.01M)||CuSO_4(1.0M)|Cu$,

the emf of this Daniell cell is E_1. When the concentration of $ZnSO_4$ is changed to 1.0 M and that of $CuSO_4$ changed to 0.01 M, the emf changes to E_2. From the followings, which one is the relationship between E_1 and E_2? (Given, RT/F = 0.059)

(1) $E_1 < E_2$

(2) $E_1 > E_2$

(3) $E_2 = 0 \neq E_1$

(4) $E_1 = E_2$

190. For a reaction between A and B the order with respect to A is 2 and the order with respect to B is 3. The concentrations of both A and B are doubled, the rate will increase by a factor of

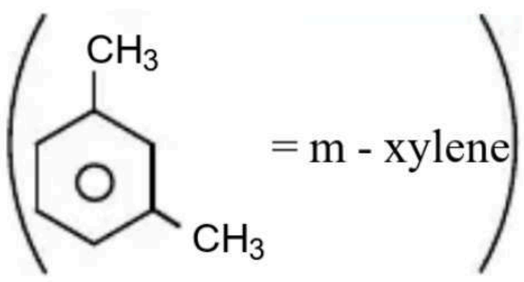
(1) 12

(2) 16

(3) 32

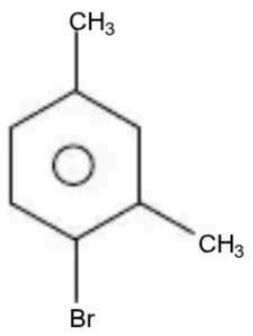
(4) 10

191. What products are formed when the m-xylene is treated with Br_2 in presence of $FeBr_3$



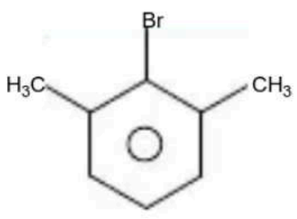
(1)



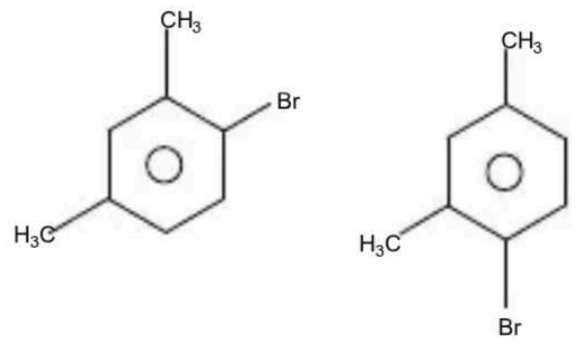


(2)

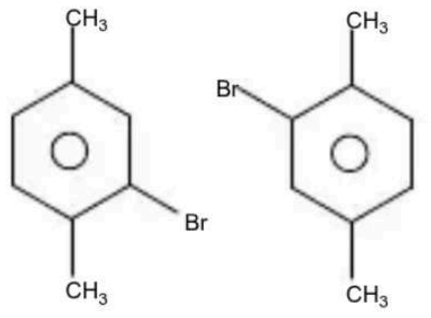




(3)



(4)



192. A cyclic hydro carbon has all the carbon and hydrogen atoms in a single plane. All the carbon carbon bonds have the same length less than 1.54Å, but more than 1.34Å. The number of mono substituted products formed by this compound is

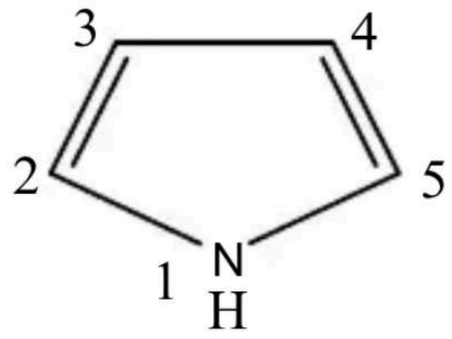
(1) 2

(2) 1

(3) 3

(4) Can't be predicated

193. In pyrrole the electron density is maximum on



(1) 2 and 3

(2) 3 and 4

(3) 2 and 4

(4) 2 and 5

194.

$$CH_3 - CH_2Cl \xrightarrow{\text{alcoholic KOH}} A \xrightarrow{Br_2} B \xrightarrow[\Delta]{Zn} C$$

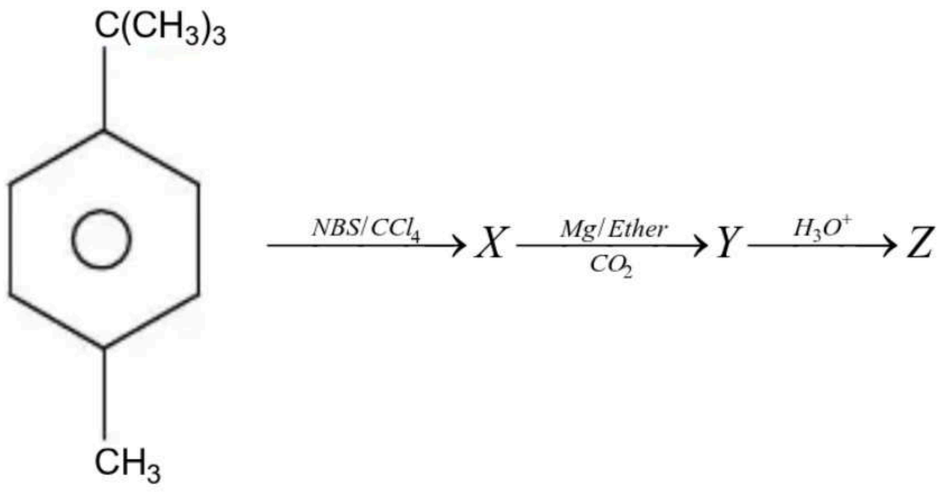
(1) Acetylene

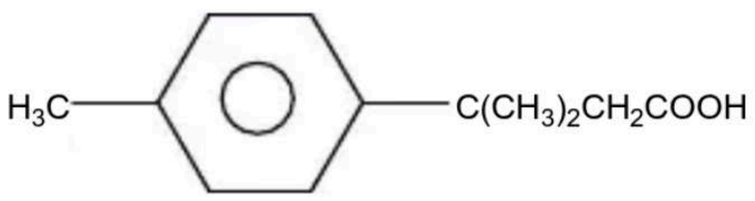
(2) Ethane

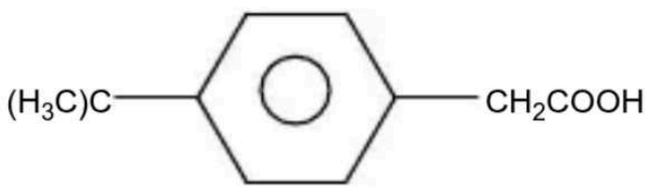
(3) Ethene

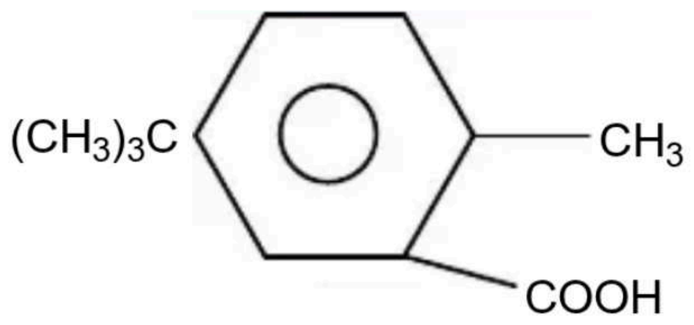
(4) Methane

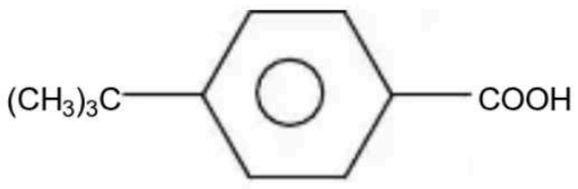
195. The final product (Z) in the following sequence of reaction is

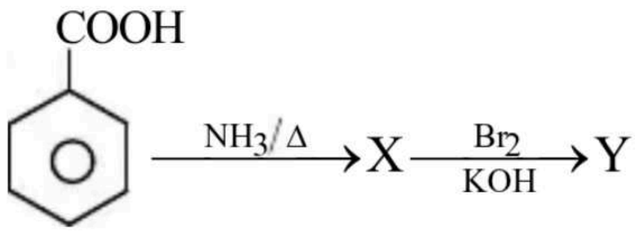


(1) 

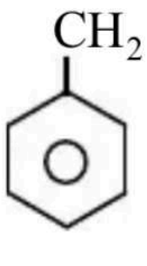
(2) 

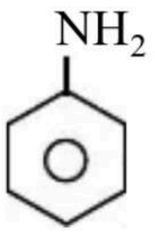
(3) 

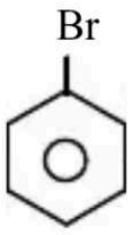
(4) 

196. 

. Y is

(1) 

(2) 

(3) 

(4) 

197. C_2H_5CHO and CH_3COCH_3 can be distinguished from one another by testing with

(1) Phenyl hydrazine

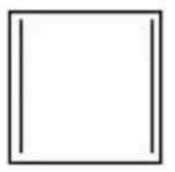
(2) 2, 4 – Dinitrophenyl hydrazine

(3) Fehling solution

(4) HCN

198. The correct order of stability of the following compound is

a) 

b) 

c) 

(1) a > c > b

(2) c > a > b

(3) a > b > c

(4) c > b > a

199. Assertion : Beryllium carbonate is kept in the atmosphere of carbon dioxide

Reason : Beryllium carbonate is unstable and decomposes to give beryllium oxide and carbon dioxide

(1) Both assertion and reason are correct and reason is the correct explanation for assertion

(2) Both assertion and reason are correct but reason is not the correct explanation for assertion

(3) Assertion is correct but reason is incorrect

(4) Both assertion and reason are incorrect

200. The reason for greater range of oxidation states in actinoid is attributed to

(1) Actinoid contraction

(2) 5f, 6d and 7s levels having comparable energies

(3) 4f and 5d levels being close in energies

(4) Progressive filling of 5f orbitals